VibeCheck: An Explainable Cross-Platform Digital Reflection and Wellbeing System

Abstract

Digital platforms such as Spotify, YouTube, and Instagram each generate rich data about a user's interests and habits, but they operate in isolation, leaving users without a unified view of their digital lifestyle. Existing recap features present statistics without context, and most recommendation systems optimize for engagement rather than user understanding or wellbeing. This project presents VibeCheck, a cross-platform system that consolidates user activity into a unified interest profile, generates personalized and shareable recap summaries, and produces explainable recommendations with clear, natural-language justifications. Beyond summarization, the system includes a digital wellbeing module that detects genuine shifts in usage patterns — using the user's own historical baseline rather than fixed thresholds — and presents them in a non-judgmental, opt-in way. The system is designed prioritizing a shareable, visually engaging recap experience alongside a private, encouraging wellbeing view.

Introduction

The rapid growth of digital platforms has changed how people learn, communicate, and consume content. Apps like Spotify, YouTube, and Instagram each hold a piece of a user's digital identity, but none of them talk to each other — so no single platform can tell a user who they've actually become over the last month. Existing "recap" features (yearly wraps, weekly summaries) show numbers without meaning, and recommendation engines are tuned to keep people scrolling, not to help them understand themselves.

VibeCheck addresses this gap by combining activity from multiple platforms into one interest profile, turning that profile into recap reports people actually want to look at and share, and pairing it with explainable recommendations — every suggestion comes with a plain-language reason. Alongside this, a wellbeing layer surfaces real behavioral

shifts (not routine noise) in a tone that's encouraging rather than clinical, leaving any action entirely up to the user.

The gap this fills: most existing research treats explainability, digital wellbeing, and cross-platform personalization as separate problems, each studied in isolation. VibeCheck is not positioned as a new explainability algorithm or a new wellbeing metric — its contribution is systems-level: unifying cross-platform interest modeling, explainable recommendation, and non-judgmental behavioral reflection into a single coherent product, built for a shareable, Gen Z-native experience rather than a dashboard.

Scope

In scope:

- Data ingestion from Spotify (live API) and YouTube/Instagram (user-exported data)
- A unified cross-platform interest profile built from that data
- Weekly/monthly shareable recap cards (visual, exportable to Instagram/WhatsApp)
- An explainable recommendation module (content/creator suggestions with a stated reason)
- A private digital wellbeing view showing time-of-day patterns, category balance, and genuine outlier detection against the user's own baseline
- A basic web application tying these features together

Out of scope (for this build):

- Full production-scale social/community infrastructure (private groups, moderation, anonymous community matching) — treated as a stretch feature or future work if time allows
- Live integrations beyond Spotify's API (YouTube/Instagram rely on user-exported data, not live API polling)

Literature Survey

#	Work	Focus	Gap relative to VibeCheck
1	Understanding Digital Wellbeing Through Smartphone Usage Intentions and Regrettable Patterns (IEEE, 2024)	Studies regrettable smartphone usage patterns and their link to wellbeing	Limited to a single device/platform; no cross-platform view, no explainability, no sharing
2	Recommendation with Dynamic Natural Language Explanations (IEEE, 2023)	Generates natural-language explanations for recommendations	Explains recommendations only — no wellbeing or reflection component
3	Explainable Recommender Integrating GNNs and User Reviews (IEEE, 2024)	Combines graph neural networks with reviews for interpretability	Single-platform interaction data; no multi-platform fusion
4	Explainable Recommendation via Sentiment Analysis + Multi-Scale Collaborative Filtering (IEEE, 2024)	Adds sentiment signals to collaborative filtering	No recap, behavior-change tracking, or privacy-aware social layer
5	KEMB-Rec: Knowledge-Enhanced Multibehavior Recommendation (IEEE, 2024)	Models multiple user behaviors with graph contrastive learning	Closest to cross-platform behavior modeling, but not framed around wellbeing or reflection
6	Review-Sentiment-Enhanced Multitask Learning for Explainable Recommendation (IEEE, 2024)	Uses review sentiment with multitask learning	No unified cross-platform lifestyle profile
7	Explainable Recommender Systems: From Theory to Practice (Springer, 2023)	Theoretical/practical foundations for explainability	Foundational only — no concrete wellbeing system

#	Work	Focus	Gap relative to VibeCheck
8	The Ethics of Digital Wellbeing: A Thematic Review (Springer, 2020)	Reviews ethical concerns in digital wellbeing design	Background/ethics only, no system proposed
9	Taxonomy of Explanations in Decision Support and Recommender Systems (Springer, 2017)	Classifies explanation types	Useful for design vocabulary, not a habit-tracking or social system
10	Privacy Protection in Cross-Platform Recommender Systems (Springer, 2023)	Explores privacy-preserving techniques for cross-platform recommenders	Close on privacy, but not focused on wellbeing reflection or recap sharing

Summary of the gap: existing work generally isolates one of three concerns — explainability, wellbeing, or cross-platform privacy — and studies it deeply within a single platform or a narrow technical scope. None of the surveyed work combines cross-platform interest fusion, explainable recommendation, and non-judgmental behavioral reflection into one user-facing system. VibeCheck's contribution sits at that intersection.

Proposed Methodology

1. Data collection

- Spotify: OAuth login via the Spotify Web API (recently played, top tracks/genres, audio features)
- YouTube: parsed from Google Takeout export (watch history, categories)
- Instagram: parsed from Instagram's data export (activity/engagement data, subject to what Instagram makes available)

2. Data processing

- Clean and normalize each platform's data into a common schema (timestamp, platform, category, item)
- Map raw categories (song genres, video categories) to a shared interest taxonomy

3. Cross-platform interest modeling

- Frequency- and recency-weighted scoring per interest category across platforms
- Optional lightweight clustering (e.g. k-means or simple embedding similarity) to detect emerging interest clusters, kept simple given team size and timeline

4. Explainable recommendation module

- Rule/template-based explanation generation initially ("recommended because your X listening rose Y% this month") — favored over a trained explainability model for feasibility within a solo final-year timeline
- Recommendation candidates drawn from category-matched content/creators

5. Wellbeing analytics

- Rolling personal baseline (mean and standard deviation of recent weeks) per user
- Outlier detection: flag a pattern only when it deviates meaningfully from the user's own baseline, not a fixed global threshold
- Insight text generated from detected deltas, written in a plain, non-judgmental tone

6. Recap card generation

- Template-based card layouts (2–4 variants) populated with top stats, one AI-generated insight line, and a discovery highlight
- Rendered as shareable images sized for Instagram Stories/WhatsApp Status (9:16)

Suggested tools/stack:

- Backend: Python (FastAPI or Flask)
- Data/ML: pandas for processing, scikit-learn for lightweight clustering, spaCy or a small HuggingFace model for category/keyword tagging if needed
- Spotify integration: `Spotify` library
- Frontend: React (or plain HTML/CSS/JS if simpler for solo development)

- Card rendering: HTML/CSS-to-image (e.g. Puppeteer/Playwright) or a Python imaging library (Pillow)
- Database: SQLite for development, PostgreSQL if scaling further

Conclusion

VibeCheck combines cross-platform interest modeling, explainable recommendations, shareable recap experiences, and gentle wellbeing reflection into a single system — an area existing research addresses only in isolated pieces. As a final year project, the scope is deliberately bounded: live Spotify integration plus export-based YouTube/Instagram data, a simple but personalized interest model, template-based explainable recommendations, and a rolling-baseline wellbeing module, all wrapped in a small web application. This keeps the build achievable within a solo, time-boxed project while still delivering a coherent, demonstrable system with a clear identity — built specifically around what makes it usable and appealing to its intended Gen Z audience, rather than treating that as an afterthought.

VibeCheck: An Explainable AI-Powered Cross-Platform Digital Reflection and Wellbeing System

ABSTRACT

The increasing use of digital platforms such as Spotify, YouTube, and Instagram has resulted in users generating large amounts of behavioral data across different applications. However, these platforms operate independently and provide only isolated summaries of user activity, making it difficult for users to gain a complete understanding of their digital habits. Existing recap systems mainly present statistics without offering meaningful insights, while recommendation systems primarily focus on maximizing engagement rather than encouraging self-reflection or digital wellbeing.

To address these limitations, this project proposes **VibeCheck**, an AI-powered cross-platform digital reflection and wellbeing system. The system collects user activity from multiple platforms, builds a unified interest profile, and generates personalized weekly or monthly recap reports. It also provides explainable recommendations by clearly describing why particular content or communities are suggested. Unlike existing systems, VibeCheck analyzes changes in user interests over time and generates AI-powered reflections that help users understand their digital behavior and promote healthier digital habits. Additionally, the platform includes privacy-preserving anonymous communities and private friend groups, enabling users to share insights securely while maintaining control over their personal data.

INTRODUCTION

Digital platforms have become an essential part of everyday life, influencing how people learn, communicate, and consume entertainment. Applications such as Spotify, YouTube, and Instagram continuously collect valuable information about user preferences, interests, and online activities. Despite this, each platform provides only its own summary, preventing users from understanding their complete digital lifestyle.

Current recap systems are limited to presenting usage statistics, while existing recommendation systems primarily focus on increasing user engagement. Very few systems help users understand how their interests evolve, explain why recommendations are generated, or encourage healthier digital habits.

VibeCheck addresses these challenges by integrating user activity from multiple platforms into a unified interest profile. Using artificial intelligence, the system generates personalized recap reports, explainable recommendations, and behavioral reflections that help users gain meaningful insights into their digital lifestyle. The platform further enhances user experience through privacy-preserving anonymous communities and private friend groups, creating a secure environment for sharing and interaction.

RESEARCH GAP

Existing research mainly focuses on individual aspects such as digital wellbeing, explainable recommendation systems, or privacy-preserving recommendations. Most existing systems operate on a single platform and do not provide a unified understanding of user behavior across multiple digital services.

There is currently no integrated system that combines:

- Cross-platform activity analysis
- Explainable AI recommendations
- Long-term behavioral reflection
- Weekly/Monthly recap generation
- Privacy-preserving community interaction

VibeCheck fills this gap by combining all these components into a single AI-powered digital reflection platform.

OBJECTIVES

- To collect user activity from multiple digital platforms.
- To build a unified cross-platform interest profile.
- To generate AI-powered weekly and monthly recap reports.
- To provide explainable personalized recommendations.
- To promote digital reflection and healthier digital habits.
- To enable secure social interaction through anonymous communities and private groups.

SCOPE OF THE PROJECT

In Scope

- Spotify API integration
- YouTube exported data
- Instagram exported data
- Cross-platform interest analysis
- AI-generated reflections
- Explainable recommendations
- Weekly and monthly recap generation
- Shareable recap cards
- Anonymous interest-based communities
- Private friend groups
- Privacy-preserving data handling

Out of Scope

- Real-time screen time monitoring
- Full Instagram and YouTube live activity APIs
- Large-scale community moderation
- Mental health diagnosis
- Commercial-scale deployment

LITERATURE SURVEY

Previous research has explored digital wellbeing, explainable recommendation systems, and privacy-preserving recommendation techniques independently.

Digital wellbeing research mainly focuses on understanding smartphone usage patterns and reducing unhealthy digital behavior. Explainable recommendation research aims to improve user trust by providing understandable explanations for recommendations. Privacy-preserving recommender systems concentrate on protecting user data during recommendation generation.

Although these studies contribute significantly to their respective domains, they generally address only one aspect of the problem. Existing recap systems provide statistics without meaningful reflection, while explainable recommender systems rarely analyze cross-platform behavior or long-term interest evolution.

VibeCheck integrates these research areas by combining cross-platform activity analysis, explainable recommendations, digital reflection, and privacy-preserving social interaction into a unified framework.

PROPOSED METHODOLOGY

The proposed system follows a multi-stage pipeline to analyze user activity and generate personalized digital reflections.

Step 1 — Data Collection

The user uploads or connects activity data from multiple platforms including Spotify, YouTube, and Instagram.

Tools

- Spotify Web API
- Google Takeout
- Instagram Data Export

Step 2 — Data Preprocessing

The collected data is cleaned, normalized, and transformed into a unified format.

Techniques

- Data Cleaning
- Data Integration
- Feature Extraction

Step 3 — Cross-Platform Interest Modeling

The system analyzes user behavior across different platforms and builds a unified interest profile.

Examples

- Music
- Gaming
- AI
- Programming
- Movies
- Fashion
- Fitness

Possible Techniques

- Sentence Transformers
- TF-IDF
- Embeddings
- Cosine Similarity

Step 4 — AI Reflection & Digital Wellbeing Analysis

The system identifies behavioral trends such as:

- Interest evolution
- Learning vs entertainment ratio
- Active usage periods
- Digital habit changes

AI generates personalized reflections to help users better understand their digital behavior.

Step 5 — Explainable Recommendation Engine

The recommendation engine suggests:

- Videos
- Music
- Podcasts
- Communities
- Learning resources

Each recommendation includes a natural language explanation describing why it was generated.

Possible Models

- Hybrid Recommendation System
- Content-Based Filtering
- Collaborative Filtering
- LLM-generated explanations

Step 6 — Weekly/Monthly Recap Generation

The platform creates visually engaging recap cards summarizing:

- Top interests
- Interest evolution
- User activity
- AI reflections
- Recommendation highlights

Users can download and share these recap cards.

Step 7 — Social Layer

The system recommends anonymous niche communities based on user interests.

Users can also create private groups with friends to compare recap reports and discover shared interests while maintaining privacy.

SYSTEM FLOW

User Uploads Data
↓
Cross-Platform Interest Analysis
↓
AI Reflection & Digital Wellbeing
↓
Explainable Recommendations
↓
Weekly / Monthly Recap
↓
Shareable Recap Cards
↓
Anonymous Communities
↓
Private Friend Groups

TOOLS & TECHNOLOGIES

Frontend

- React.js
- Tailwind CSS

Backend

- FastAPI

Database

- MongoDB

AI & Machine Learning

- Python
- Hugging Face Transformers
- Sentence Transformers
- Scikit-learn

APIs

- Spotify Web API
- Google Takeout
- Instagram Data Export

Visualization

- Plotly
- Chart.js

Deployment

- Docker
- Render / Railway / Vercel

EXPECTED OUTCOME

The proposed system will enable users to understand their digital lifestyle through AI-generated reflections instead of simple statistics. It will provide personalized recap reports, explainable recommendations, secure social sharing, and anonymous community participation while preserving user privacy. The system is expected to improve user awareness of digital habits and support healthier digital engagement.

CONCLUSION

VibeCheck presents an AI-powered cross-platform digital reflection and wellbeing system that combines user activity from multiple platforms into a unified interest profile. Unlike traditional recap applications or recommendation systems, it provides

explainable recommendations, AI-generated behavioral reflections, and privacy-preserving community interaction within a single platform.

By integrating digital reflection, explainable artificial intelligence, and secure social features, the proposed system offers a novel approach to helping users understand their evolving digital habits rather than simply tracking online activity. This combination makes VibeCheck a meaningful contribution to the areas of digital wellbeing, explainable recommendation systems, and cross-platform user behavior analysis while remaining feasible for implementation as an MSc-level research project.